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REVIEW

# Pain management after thoracic surgery or chest trauma

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ABSTRACT

Accidental or surgically induced thoracic trauma is responsible for significant pain that can impact patient outcomes. One of the main objectives of its pain management is to promote effective coughing and early mobilization to reduce atelectasis and ventilation disorders induced by pulmonary contusion. The incidence of chronic pain can affect more than 35% of patients after both thoracotomy and thoracoscopy as well as after chest trauma. As the severity of acute pain is associated with the incidence of chronic pain, early and effective pain management is very important. In this narrative review, we propose to detail systemic and regional analgesia techniques to minimize postoperative pain, while reducing transitional pain, surgical stress response and opioid side effects. We provide the reader with practical recommendations based on both literature and clinical practice experience in a referral level III thoracic trauma center.

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KEY WORDS: Thoracic injuries; Thoracic surgery; Pain management; Conduction anesthesia.

## Chest pain control: a major issue in both thoracic surgery and chest trauma

### Thoracic surgery

Thoracic surgery, by thoracotomy, videothoracoscopy (VATS) and robot-assisted thoracic surgery (RATS), is one of the most pain-producing procedures, whether that be acute or chronic.[1] Chest pain leads to hypoventilation and inhibits effective coughing, which favors the development of atelectasis, bronchial stasis and postoperative pneumonia. The positive impact of pain control in postoperative thoracic surgery on the short- and long-term outcome as well as on the quality of life has been well demonstrated.[2]

Enhanced recovery after thoracic surgery is now established.[3] One of the key measures of this early rehabilitation is pain management in order to allow, among other things, the resumption of walking and physical therapy.

### Thoracic trauma

Chest trauma accounts for one-third of all trauma cases and may lead to fatal complications. Rib fractures, especially flail chest is a severe thoracic injury associated with significant morbidity and mortality. Patients may develop severe acute respiratory failure.

Age, pre-existing respiratory or cardiac comorbidity, number of fractured ribs and pulmonary contusion are recognized factors of severity of pulmonary and systemic complications. The Thoracic Trauma Severity Score (TTSS) has been developed for predicting the occurrence of

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acute respiratory distress syndrome (ARDS) and mortality in patients with chest trauma (Supplementary Digital Material 1: Supplementary Table I). This score includes: age of patients, the $PaO_2/FiO_2$ ratio, the number of fractured ribs, and the existence of pulmonary contusion and pleural involvement on computed tomography (CT)-scan.

Broken ribs, by intercostal, pleural and muscular damages cause acute pain and potentially lead to chronic pain. Surgical fixation is sometimes proposed increasing the risk of chronic pain and, then, respiratory impairment particularly in older patients. The consequences of acute pain after chest trauma on respiratory function are well described in the literature.[4] Poor pain control can lead to respiratory distress with the need for mechanical ventilation.

Rapid pain control is therefore essential. The experts suggest assessing pain at rest and during exercise (effective cough and deep inspiration) with a target of visual analogical score less than or equal to two. Regional analgesia should be performed for patients with factors of severity for pulmonary complications and for patients with uncontrolled pain in the first 12 hours.[5] The Pain, Inspiratory effort, Cough (PIC) score is used to assess pain at rest but also during inspiration and coughing. Its simplicity allows for replication of the assessment for monitoring the patient's evolution. If the total score is ≤4 or 1 in any of the categories, analgesia should be reinforced (Supplementary Digital Material 2: Supplementary Text File 1).

### A common goal

During thoracic surgery, as after thoracic trauma, the challenge is the same: to allow the patient to cough effectively and mobilize quickly in order to reduce atelectasis and ventilation disorders induced by pulmonary contusion and injuries (Figure 1).

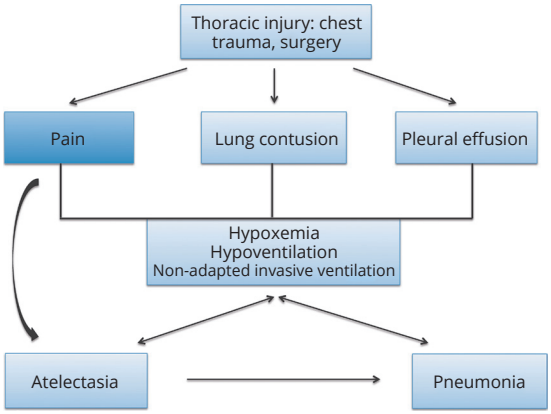



Figure 1.—Pathophysiological mechanisms of respiratory complications in patients with acute chest pain.

## Pathophysiology of thoracic pain

Postoperative pain is both nociceptive and neuropathic and comes from somatic and visceral afferents (Figure 2).[6, 7] Intercostal nerves and their branches carry somatic nociception. Direct lesion produces nociception, through skin incision, rib spreading or fracture, stretching of the wall muscles (latissimus dorsi, intercostal, trapezius and serratus anterior muscles), pleura effraction and chest drain insertion. Then, the tissue injury leads to a release of inflammatory mediators that directly activate the nociceptors and reduce the pain threshold. Lesions of the bronchi, visceral pleura or pericardium produce visceral nociception carried by the phrenic and vagus nerves.

Neuropathic pain is, mainly due to nerve injury or related to care. Patients factors, intraoperative manipulations and the perioperative position of the patient are likely to participate in the process of peripheral and central sensitization, in particular *via* the dorsal nerve of the scapula, long thoracic nerve and thoraco-dorsal.

Neuropathic pain occurs in more than 29% of thoracotomy patients and can reach 35% after thoracoscopy.[8] Kahloul *et al*. found that 79.6% of patients experienced chronic pain after chest trauma.[9] It varies according to procedure, physical and mental health, preoperative pain, the severity of the tissue trauma, and the timing of the analgesia. Since the severity of acute pain is associated with the incidence of chronic pain after thoracic surgery, efficient pain management is of the utmost importance.[10]

## Systemic analgesia: necessary but insufficient

### Basic analgesic

Multimodal analgesia should include, whenever possible, non-steroidal anti-inflammatory drugs (NSAIDs) which are effective on mechanical and

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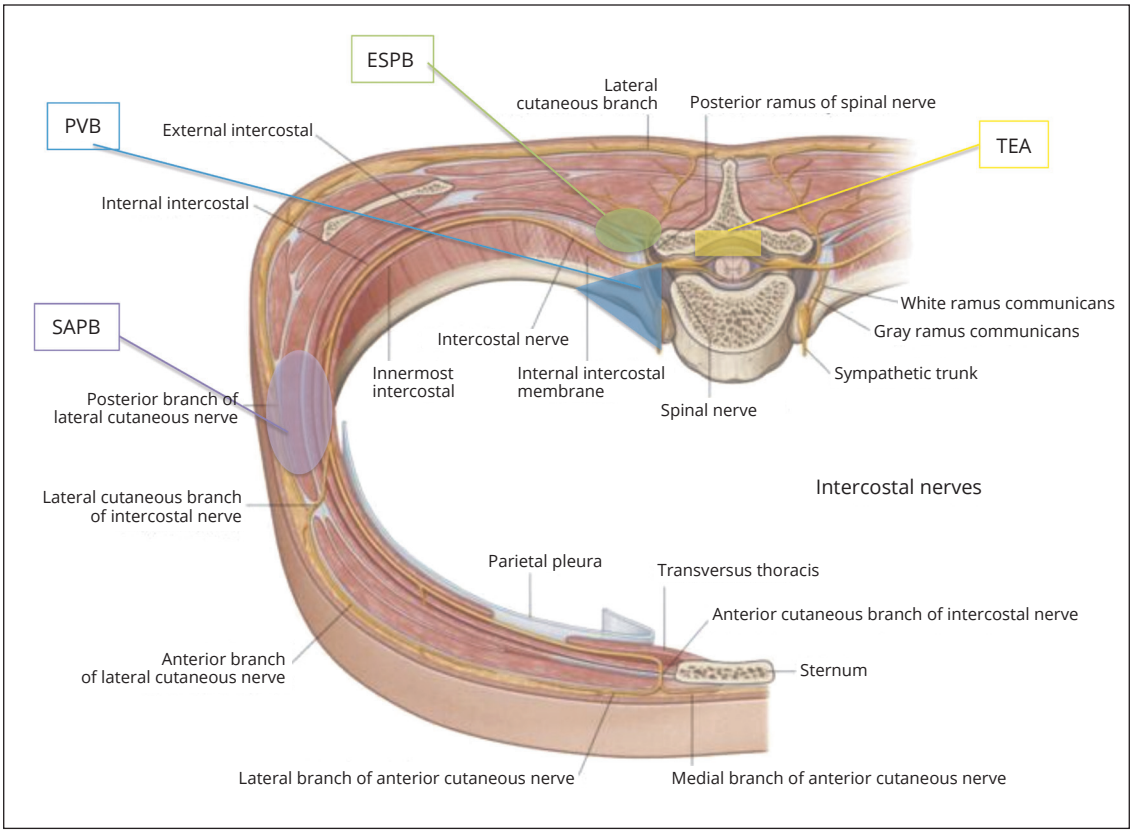



Figure 2.—Representation of the main thoracic blocks.[6]

inflammatory pain.[11] NSAID possess the highest opioid-sparing effect among systemic analgesic in the preoperative period. They can also reduce the intensity of ipsilateral shoulder pain induced by pleural and pericardial neural input and which is common after thoracotomy. Intravenous Parecoxib associated with a paravertebral block reduced acute pain scores at rest and coughing, activation of pro-inflammatory cytokines and the restrictive ventilatory profile after VATS.[12] The use of celecoxib for 48 h in combination with epidural analgesia was associated with a reduction in pain scores at rest, during coughing and mobilization without difference in postoperative pulmonary complication.[13]

Few procedure-related studies demonstrated the effectiveness of IV or oral acetaminophen, but its morphine sparing effect associated to a safe tolerance profile motivates its systematic use.[14] Paracetamol (4 g per day) was as effective as ketorolac (120 mg) on pain scores in patients scheduled for VATS without regional analgesia when compared with intravenous morphine (despite lower opioid rescue analgesia in the ketorolac group).[15]

### NMDA antagonists

The hyperexcitability of nociceptive information from spinal and suprarenal structures activates NMDA-receptors worsening the risk of neuropathic pain. Ketamine reduces the intensity of pain and decreases the postoperative consumption of morphine.[16] It can even improve the effects of the epidural.[17] However, a recent study failed to demonstrate a benefit of ketamine on the reduction of chronic pain.[18] Even if there is no strong evidence of its benefit in videothoracoscopy or in thoracic trauma, its ease of administration and few reported side effects lead us to expand its use in all major surgeries. In a data-

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base including more than 80,000 patients operated between 2009 and 2018 in 607 hospitals, ketamine use was increased in open surgery, VATS and RATS despite showing little or no benefice. In this cohort, despite its effectiveness, epidural analgesia was reduced by 50%, the use of anti-NMDA agent having been promoted to compensate for this reduction.[19] We encourage clinicians to use ketamine as an adjunct to regional anesthesia or while waiting for it to be performed.

### Intravenous alpha-2 adrenergic agonists

As an alternative to locoregional anesthesia, alpha-2 adrenergic agonists could have their place in thoracic surgery because they allow a decrease in postoperative pain scores and morphine consumption.[20] However, the benefits of alpha-2 adrenergic agonists during the postoperative period has not been demonstrated. The safety profile is narrow which does not encourage systematic use in clinical practice, particularly outside an intensive care unit.[21] Using the Surgical Pleth Index (SPI) as nociceptive indicator, a placebo-controlled study found that patients receiving a preoperative bolus of dexmedetomidine 0.8 µg/kg had reduced pain scores and SPI at the PACU discharge without differences during the 48 postoperative hours.[22] In a propensity-score matched analysis, the combination of dexmedetomidine and ketamine failed to reduce pain scores and opioid requirement or postoperative complications.[23]

Alpha-2 adrenergic agonists could be beneficial for patients at high risk of confusion or postoperative agitation. Dexmedetomidine 1µg/kg administered preoperatively improved QoR-40 scores, a questionnaire that measures the overall quality of recovery after surgery, registered on postoperative day one and two.[24] It also reduced emergence agitation after VATS when compared with placebo without differences in postoperative complications.[25]

### Intravenous lidocaine

In a placebo-controlled study, intravenous lidocaine infusion has not shown to be effective in the management of thoracic trauma.[26] Thoracic paravertebral block seems to be more effective in providing analgesia than intravenous lidocaine in thoracic surgery.[27]

An international consensus on efficacy and safety of intravenous lidocaine has stated that "Intravenous lidocaine should not be used at the same time as, or within the period of action of other local anesthetic interventions"[27]. A recent propensity-score study found clinical relevant differences 24 hours and 48 hours after VATS or RATS in patients who received an opioid-sparing protocol including the combination of IV lidocaine, ketamine and dexmedetomidine but, again, without benefit on pulmonary complications.[28]

We therefore advise against the use of intravenous lidocaine for chest analgesia, especially if a thoracic block is performed.

### Steroids

There is no evidence that the use of corticosteroids is advantageous specifically in the analgesia of thoracic trauma or thoracic surgery.[29] Dexamethasone at induction has been shown to reduce throat pain after double-lumen intubation.[30] Its benefit in multimodal analgesia, in preventing nausea and vomiting and its low incidence of side effects encourages its use in induction in thoracic surgery. A recent RCT found that IV dexamethasone 8 mg combined to a bi-level single shot paravertebral block significantly delayed the time to first analgesic requirement by 35 minutes during VATS when compared to a paravertebral block alone, and reduced postoperative morphine consumption at 6, 12, 24 and 48 hours.[31] A Cochrane's meta-analysis did not find any increase in postoperative infection with a single dose of IV dexamethasone in 4603 patients but a significantly increased glucose values in the first 12 hours after surgery with a mean difference of 13.31 mg/dL.[32]

Given the lack of studies of sufficient methodological quality, consequently, corticosteroid cannot be proposed in the management of thoracic trauma.

### Gabapentinoids

Thoracic surgery and chest trauma are sources of neuropathic pain. Given the anti-allodynic properties of gabapentinoids, perioperative use of gabapentin or pregabalin was proposed to improve immediate postoperative pain control and for the

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prevention of persistent post-thoracotomy pain. However, gabapentin does not appear to have any prophylactic value in reducing VAS scores or preventing pain chronicization. In a propensity-matched analysis including more than 70,000 patients in open and video thoracic surgeries, gabapentin was associated with higher postoperative respiratory failure requiring non-invasive and invasive ventilation. This molecule is to be reserved for the curative treatment of neuropathic chronic pain especially after chest trauma.[33] The latest French expert recommendation places gabapentin as the first line of treatment for neuropathic pain.[34]

### "Opioid-based anesthesia" and "opioid free anesthesia" (OFA)

Due to the intensity of chest wall pain, opioids are frequently required.[19] No particular type of opioid can be recommended and practices vary widely around the world. However, the current trend is to reduce morphine use to reduce its side effects (hyperalgesia, chronic postoperative pain, respiratory depression, urinary retention, slowing transit, postoperative nausea and vomiting and postoperative delirium) as well as pulmonary functions. Finally, although not proven, opioids possess a theorical potential deleterious impact on lung cancer progression.[35] Studies on OFA in thoracic surgery are limited. Regional anesthesia, such as the paravertebral block, remains one of the most effective technique for OFA with effective control of nociception.[36] There is no high-level evidence that OFA reduces postoperative morphine consumption in thoracic surgery. Balanced OFA using continuous infusion of dexmedetomidine 0.4 to 1.4 μg/kg/h was associated with severe bradycardia in a recent randomized multicenter study leading to its premature termination.[21] Other results were not in favor of dexmedetomidine with more postoperative hypoxemia, delayed extubation, prolonged PACU stay,[21] However, adequately dosed OFA, particularly in absence of regional analgesia, can be proposed effectively to reduce the first 48 hours postoperative pain scores.[28] Finally, the use of locoregional anesthesia seems to be the best way to reduce morphine consumption and its negative side effects.

## Regional analgesia: from central to peripheral nerve block

### Thoracic epidural analgesia

Thoracic epidural analgesia (TEA) provided greater analgesia and less opioid consumption when compared to systemic analgesia alone.[37] However, thoracic epidural catheterization remained sometimes challenging even in experienced teams with, as main identified risks of failure, the upward angulation of the Tuohy needle and the BMI. The failure rate for thoracic epidural catheter placement can range from 12% to 40% depending on the cohort.[38]

TEA is more invasive than parietal blocks. It conveys risks of hypotension, urinary retention and potential lower limb weakness, which can delay early rehabilitation and resumption of walking.

Thoracic epidural remains the optimal technique for:

- lung resection with pleural decortication, thoracotomy;
- bilateral chest trauma or unilateral chest trauma with more than six broken ribs or high risk of complications;
- when paravertebral block is not feasible (poor ultrasound visualization).

### Paravertebral and midpoint transverse process-to-pleura blocks

The paravertebral block (PVB) has been shown to be effective compared to systemic analgesia alone in many randomized and systematic analysis.[39-41] Thoracic PVB provides similar analgesia as TEA with lower incidence of perioperative hypotension, nausea and vomiting, and urinary retention.[42, 43]

The failure rate of the paravertebral block is slightly lower than TEA ranging from 6% to 15%. The catheter can be placed by the surgeon under direct visualization.

However, in case of pleural damage or large pleural resection, a catheter cannot be inserted with a risk of hazardous diffusion of local anesthetic. The midpoint transverse process-to-pleura (MTP) technique was developed as a safer alternative to PVB.[44] The tip of the needle is placed at

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the mid-point between the transverse process and pleura behind the superior costotransverse ligament. The anterior displacement of the pleura is inconsistently observed during the injection of local anesthetic which can reach the paravertebral space using superior costotransverse ligament fenestration and the costotransverse junction. In a randomized trial, a midpoint process-to-pleura performed at a single level was compared with a PVB during VATS with pain scores at rest as primary outcome.[45] The pain score at 12 hours was similar in the two groups as well as the time of the first morphine requirement but the MTP procedure was carried out more quickly. This technique can also be proposed during the management of a chest tube insertion.

The main limitation of the PVB is the difficulty to correctly identify the ultrasound landmarks, particularly the pleura and its anterior displacement when the tip of the needle reaches the anterior part of the superior costotransverse ligament, with the risk of pleura puncture. The difficulty of ultrasound visualization increases with obesity or emphysema. Other complications are block failure (5-10%), hypotension and vascular puncture (<3%). The paravertebral space can be identified using a transverse orientation with a classically lateral-to-medial needle orientation, or a sagittal view with a caudal-to-cephalic needle pathway. In the lateral-to-medial orientation, more epidural spread is observed.

During VATS, a single-level PVB performed before surgery was more effective than an intercostal nerve block administered by the surgeon at the end of the surgery in pain scores at rest, during coughing and in opioid rescue.[46] A recent randomized study compared a multi-level (T5-T7) PVB to a T4-T9 intercostal block and an ESPB performed at T5 using 20 mL of ropivacaine 0.375%.[47] The cumulative morphine consumption was significantly less in the PVB when compared with the intercostal nerve block and the ESPB. The VAS scores at rest and coughing during the first 8 postoperative hours were also lower with PVB.

Paravertebral block is the preferred technique for:

• lung surgery without decortication (thoracotomy or VATS);

• unilateral chest trauma if the visualization of the paravertebral space is correct (and when a chest tube is already in place).

### Erector spinae plane block

The Erector spinae plane block (ESPB) is a more recent block. It is performed under ultrasound guidance, with a simple visualization of the transverse process making this block easy to perform. It also carries less risk of pleural puncture, which is advantageous in the context of thoracic trauma in a patient without a chest tube.

However, its effectiveness is debated. The exact mechanism of action remains undefined and the clinical effect unpredictable. Its popularity is clearly based on a more easily identifiable sonoanatomy than for a BPV or a TEA. Local anesthetic injected during an ESPB involves the posterior rami of the spinal nerves and diffuses posteriorly in the erector spinae fascial plane, the trapezius, latissimus dorsi, rhomboid, serratus posterior and splenius muscles. Extension toward the paravertebral space remains controversial despite the medial and lateral porosity of the superior costotransverse ligament and muscular pathway within the posterior rami of the intercostal nerves and vessels or through the intercostal membrane.[48] During an ESPB, the lateral extension is frequently observed and participates in the analgesic parietal component during thoracic surgery reinforcing the role of the anatomical and functional continuity between the anterior serratus, the rhomboids and levator scapulae muscles. Other proposed mechanisms include systemic resorption, immunomodulatory effect and lymphatic involvement but require further investigation. There is little clinical data confirming a multilevel extension toward the sympathetic chain and the anterior spinal ramus during thoracic surgery, these two targets the remaining two key structures in pain modulation after chest trauma or thoracic surgery. The spread of local anesthetics after ESPB has been studied in healthy volunteers using MRI.[49] The authors found that local anesthetic consistently spreads to the dorsal rami of the spinal nerve, through the intertransverse connective tissue complex into the paravertebral space and to the neural foramina. Spread to the epidural space was variable. The number of anes-

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thetised dermatomes appears to be greater with PVB than with ESPB. This study also shows that sensory loss does necessarily correlate to the actual spread of local anesthetics, and therefore probably cannot be relied upon to assume clinical effect.

Some studies have shown efficacy of the erector spinae block in lung surgery[50-52] compared to systemic analgesia or compared to TEA and to PVB. A non-inferiority study compared a continuous PVB and ESPB during VATS with the median difference in NRS scores at rest 24 hours postoperatively. ESPB appeared non-inferior and no difference was detected in the PCA fentanyl consumption. However, the sensitive dermatomal spread was non circumferential in the ESPB, with less parasternal sensory block, and more additional analgesic was required.[53] This RCT confirms that the mechanism of analgesia after ESPB remains a subject of debate. An ESPB was superior to a SPB in a recent RCT after posterolateral thoracotomy and was similar to PVB for pain scores and opioid requirement.[54] Conversely, an ESPB performed at the T5 level did not improve QoR-40 on postoperative day one and did not reduce the median 24 hours morphine equivalent use when compared with patients randomised in the placebo group.[55] There is no prospective study in the context of thoracic trauma but several retrospective studies have shown its efficacy to reduce pain and opioid consumption.

To target wall and sympathetic tissue analgesia at the same time, TEA and PVBs are those with the best degree of evidence.

Erector spinae plane block is the preferred technique for:

- unilateral chest trauma if visualization of the paravertebral space is difficult or if a chest tube is not in place to avoid the risk of pneumothorax;
- lung surgery with pleural decortication as an alternative when landmark or ultrasound repairs for PVB or epidural puncture are difficult.

### Serratus plane block

The serratus anterior muscle covers the anterolateral aspect of intercostal muscles from the first to the eighth ribs. The Serratus plane block (SPB) is easy to perform and the risks are almost zero. Local anesthetic can be injected between the intercostal muscle and the serratus muscle (deep serratus plane block) or more superficially between the serratus muscle and the latissimus dorsi (superficial serratus plane block). The serratus plane block reaches the lateral cutaneous branch of T2-T6 intercostal nerves, the long thoracic and thoracodorsal nerves. The sub-serratus injection can also reach the intercostal nerves. Insertion of a catheter between the fascia of the muscles to allow continuous injection is possible but not easy. There is no randomised controlled study of continuous administration of the serratus block.

A review comparing serratus plane block to systemic analgesia alone[56] shows that this block allows a diminution of pain scores and morphine consumption.

Studies comparing the serratus plane block with other fascial planes or neuraxial blocks seem to find a better efficacy of ESPB[51, 57, 58] while serratus plane block was found to be non-inferior when compared to a paravertebral approach.[59]

Serratus plane block is the preferred technique for:

- unilateral chest trauma when the fractures are located on the lateral or anterior portion of the ribs;
- VATS lung surgery for uniport procedures (wedge, biopsy) or in second intention (after PVB or ESPB).

### Pecto-intercostal and transverse thoracic blocks

The innervation of the area between the midclavicular line and the sternum is derived from the T2-6 intercostal nerves which travel in the intercostal space bounded by internal and innermost intercostal muscles. The intercostal nerve runs between the innermost and inner intercostal muscles. Near the sternum, it runs between the transverse thoracic and the internal intercostal muscle in the same plane as the internal thoracic artery. After going through the internal intercostal muscle and external intercostal membrane anteriorly, the terminal anterior cutaneous branch of the nerve ascends through the intercostal and pectoralis major muscles to innervate the superficial tissues of the parasternal region. This terminal branch can be blocked deep between the

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pectoralis major and internal intercostal muscles (the pecto-intercostal facial plane); or between the internal intercostal muscles and the transversus thoracis muscles (transverse thoracic muscle plane). Despite limited RCTs on these two blocks in thoracic surgery and chest trauma, anterior rib and sternal fractures can benefit from these two anterior approaches, particularly in addition to another thoracic fascial plane block such as the serratus plane block, the rhomboid-intercostal block or even the erector spinae plane block where anesthetic extension to the anterior branch of the spinal nerve is rarely observed.

### Intercostal nerve block and pleural blocks

The different chest wall blocks have shown their superiority over the intercostal nerve block (ICNB).[60] This is probably due to the limited diffusion of local anesthetic to the intercostal nerve only, requiring multiple injections to provide analgesia. Recently, a meta-analysis including 66 RCT and observational studies compared intercostal nerve block, systemic analgesia and regional analgesia techniques. ICNB reduces static and dynamic pain scores after thoracotomy during the first 48 hours postoperative hours but only during the first six hours after VATS. ICNB was, however, inferior to epidural analgesia and PVB for pain scores and opioid consumption. This meta-analysis suffered from a high risk of bias and incomplete data reducing the quality of evidence of the qualitative analysis. There was no convincing data in all RCTs to detect a clinical impact on ICNB during chest trauma and its potential benefit in preventing chronic pain.

### Phrenic nerve block for ipsilateral pain shoulder in thoracotomy

After thoracotomy, half of the patients experience clinically relevant shoulder pain on POD 0, about 20% on POD 1 and 15% on POD 2. Pain can last up to 12 months after surgery.[61]

Several studies have looked at the phrenic nerve block. Local anesthetic is injected close to the diaphragm into the peripheric fat pad. It seems that phrenic nerve infiltration significantly reduced the incidence of thoracic and shoulder pain and delayed its occurrence compared to systemic analgesia (with acetaminophen and NSAIDs) alone without complications such as nerve injury or respiratory compromise.[62]

### Transcutaneous lidocaine

The preventive skin analgesia with transcutaneous lidocaine patch 5% seems to be effective in reducing post thoracotomy pain, at rest and after coughing, and total morphine consumption.[63] In thoracic trauma patients, on the other hand, the use of patches is of little benefit.[64]

### The place of adjuvants

To prolong the duration and depth of the analgesic properties, several adjuvants have been administered in addition to local anesthetic solutions. When single injection locoregional analgesia is performed, the use of dexmedetomidine seems to prolong analgesia compared to ropivacaine alone.[65] However, there is no evidence that the adjunction of dexmedetomidine reduces chronic pain.

It remains to be demonstrated whether a comparable effect could be achieved with I.V. administration of dexmedetomidine making it more appropriate for clinical use.

The level of evidence for perineural dexamethasone is low when compared with a systemic administration. The interest of adding sufentanil to the local anesthetic in epidural are well demonstrated. It has no interest in paravertebral blocks and other blocks since there are no opioid receptors present.

### What is the place of non-drug techniques?

The transcutaneous electrical nerve stimulation (TENS) was tested in thoracic surgery. However, studies suffered from small sample size and with methodological insufficiency leading to controversy over application in daily practice. Moreover, no randomized trials have compared TENS to multimodal analgesia including regional analgesia.

A review on the use of acupuncture[66] in thoracotomy found insufficient data to conclude on the efficacy of this analgesic technique. There is insufficient evidence to recommend cryoanalgesia.

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### Surgical techniques: does it change anything about pain?

It has been suggested that there may be a reduction in pain depending on the approach or surgical technique. A study comparing pain after open thoracotomy, VATS and RATS did not find any difference between the three different modalities.[67] Surgical techniques which preserve the intercostal neurovascular bundle do not seem to reduce acute and chronic pain after thoracotomy.[68] Strategies comparing surgical rib fixation treatment (SRFF) with medical treatment found no difference in terms of acute or chronic pain. Single port thoracoscopy seems to cause less postoperative pain and less chronic pain than three port thoracoscopy.[69] The chest tube is responsible for pain: its early removal is therefore one of the main ways to reduce postoperative pain.

## Conclusions

Thoracic wall lesion — due to trauma or induced by surgery — is responsible for clinically significant acute pain which conditions the prognosis of the patients. The development of chronic pain is frequent. Pain management must therefore include systemic analgesia and regional anesthesia, whenever possible, as early as possible.

Systemic analgesia is essential and the first step to relieve the patient as quickly as possible.

The two most effective and proven methods of analgesia are the thoracic epidural and the paravertebral block. Erector spinae plane block can mime inconstantly a similar extension to the paravertebral block to the anterior and posterior spinal arches. Nevertheless, it requires direct comparative studies before it can be recommended in these clinical situations.

Serratus plane block is easy to achieve but it is anatomically incomplete and should probably be reserved for minimally invasive procedures. Further studies are needed to better assess their effectiveness in different clinical situations. There is no evidence that VATS or robot-assisted surgery result in less pain. We suggest two schemas to guide the clinician in the most appropriate analgesic strategy for each context, whether in thoracic surgery or in the thoracic trauma patient (Figure 3, 4).

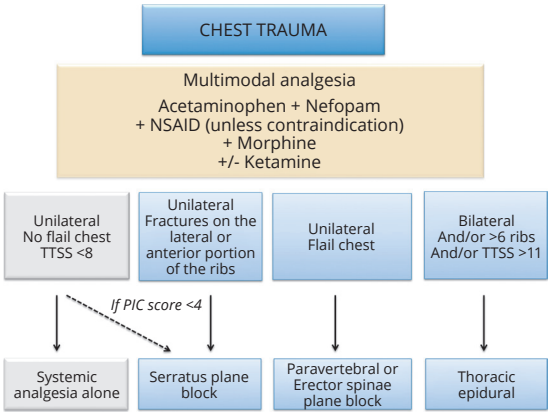



Figure 3.—Suggested analgesic strategy for the thoracic trauma patient.

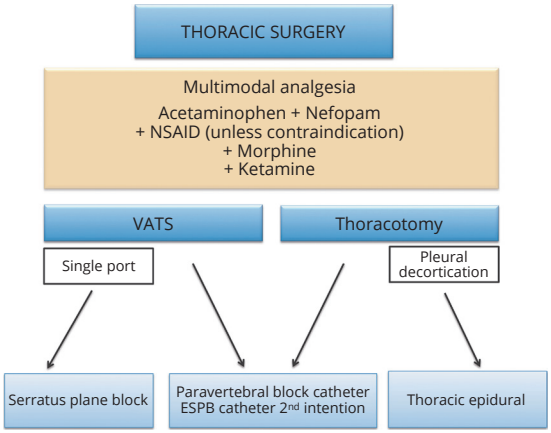



Figure 4.—Suggested analgesic strategy for thoracic surgery.

### Key messages

- Thoracic surgery is one of the most pain-producing procedures. The pathophysiology of pain in chest trauma is similar to scheduled thoracic surgery. Chronic pain can affect more than half of patients after trauma or elective thoracic surgery. Chest pain leads to hypoventilation and inhibits effective coughing, which favors the development of atelectasis, bronchial stasis and postoperative pneumonia. Early and effective pain management is therefore essential.
- Regional analgesia must be performed in addition to systemic multimodal analgesia. The two most established techniques are

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epidural and paravertebral blocks. Erector spinae and serratus plane blocks need to prove their effectiveness in different clinical situations.

- The analgesic strategy must be tailored to the context taking into consideration feasibility, efficacy and safety.

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*Conflicts of interest*
The authors certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

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